

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

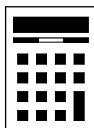
Paper 1

Monday 4 November 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



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ANSWER IN THE SPACES PROVIDED**



Answer **all** questions in the spaces provided.

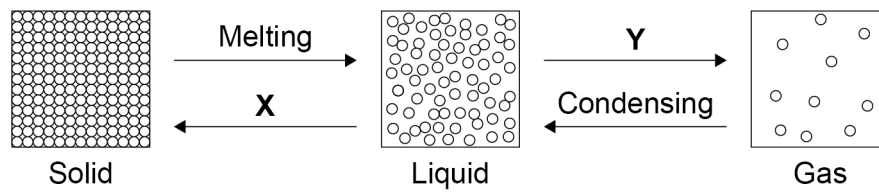
0 1

The three states of matter are solid, liquid and gas.

0 1 . 1

Figure 1 represents the three states of matter.

Figure 1



X and **Y** represent changes of state.

Name change of state **X** and change of state **Y**.

[2 marks]

X _____

Y _____

Question 1 continues on the next page

Turn over ►



Substance **A** is a solid at room temperature.

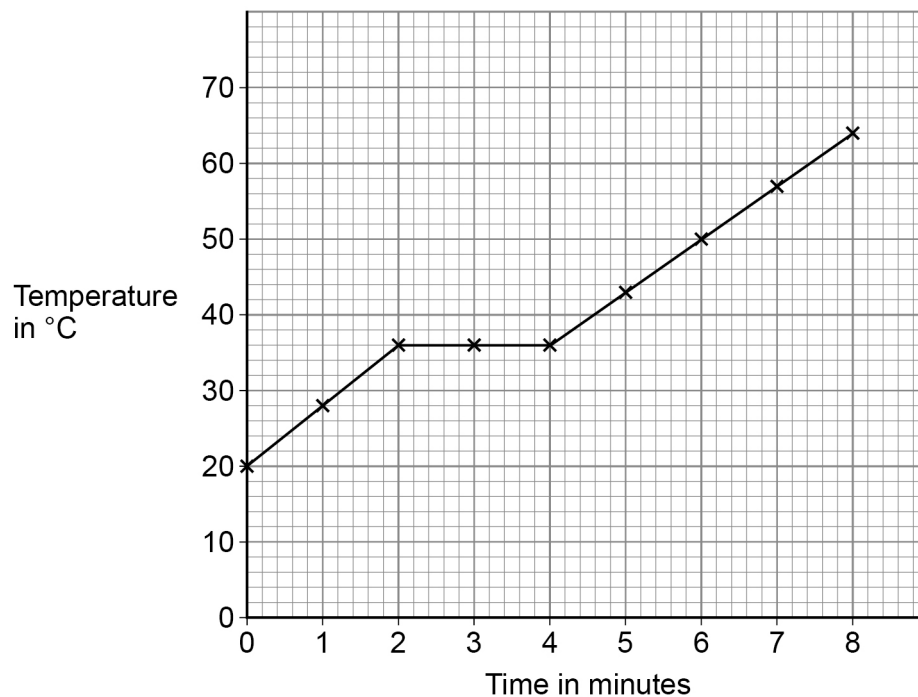
A student investigated the effect of heating substance **A**.

This is the method used.

- 1 Measure the initial temperature of substance **A**.
- 2 Heat substance **A**.
- 3 Measure the temperature of substance **A** every minute for 8 minutes.

Figure 2 shows the results.

Figure 2



0 1 . 2 Describe how the temperature of substance **A** changes during the investigation.

Use **Figure 2**.

[3 marks]



0 1 . 3 What is the melting point of substance **A**?

Use **Figure 2**.

[1 mark]

Tick (✓) **one** box.

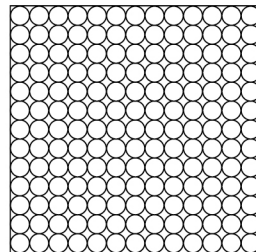
20 °C ☐

36 °C ☐

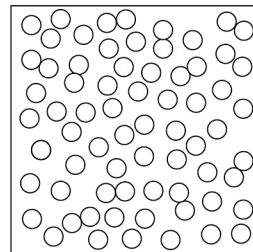
64 °C ☐

0 1 . 4 **Figure 3** shows the arrangement of particles in substance **A** at 1 minute and at 6 minutes.

Figure 3



At 1 minute



At 6 minutes

Give **one** difference between the arrangement of particles in substance **A** at 1 minute and at 6 minutes.

[1 mark]

Question 1 continues on the next page

Turn over ►



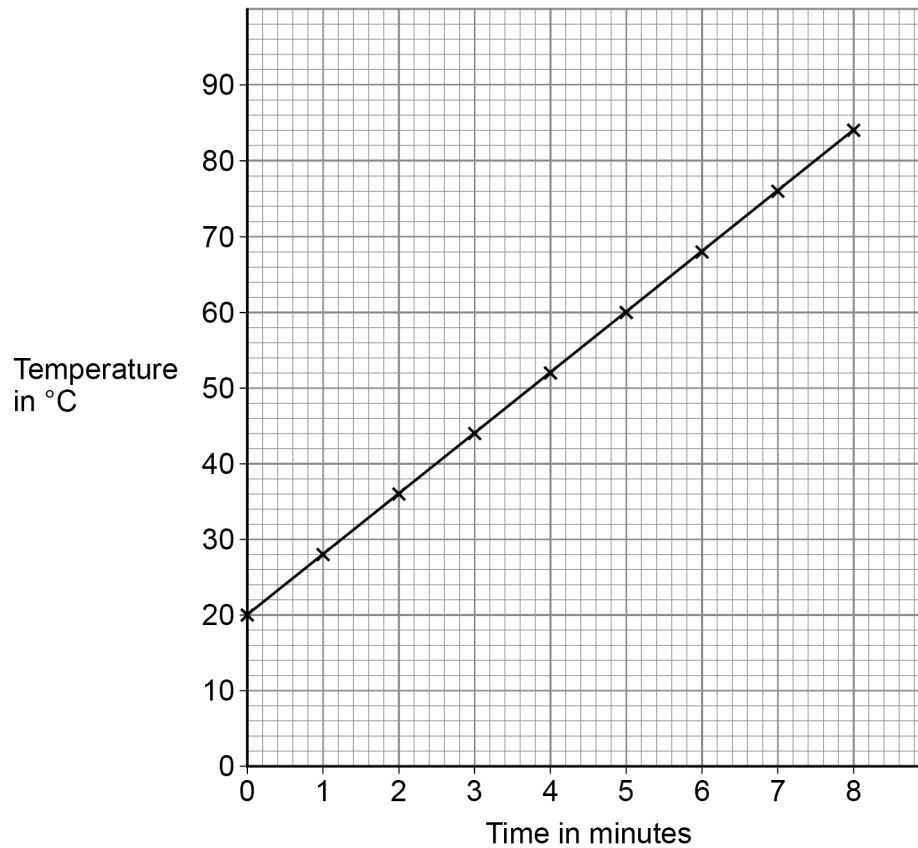
0 1 . 5

A student repeated the investigation with substance **B**.

Substance **B** is also a solid at room temperature.

Figure 4 shows the results.

Figure 4



Suggest **one** reason why substance **B** did not change state.

[1 mark]



Turn over for the next question

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0 2

This question is about the element neon.

0 2 . 1

Neon is in Group 0 of the periodic table.

What name is given to Group 0?

[1 mark]Tick (✓) **one** box.

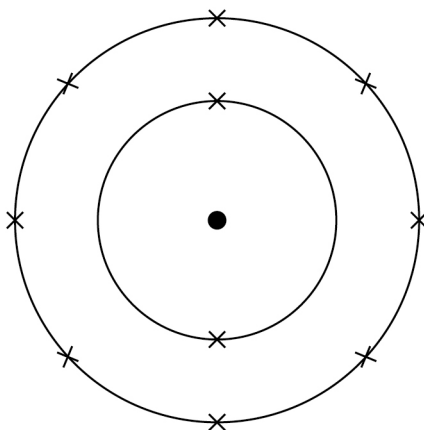
Alkali metals

☐

Halogens

☐

Noble gases

☐**Figure 5** represents a neon atom.**Figure 5****0 2 . 2**

What name is given to the centre of the atom?

[1 mark]



0 2 . 3 Neon atoms contain sub-atomic particles with different charges.

Draw **one** line from each sub-atomic particle to the correct charge.

[2 marks]

Sub-atomic particle

Charge

Electron

+1

Neutron

0

Proton

-1

0 2 . 4 Which sub-atomic particle has the lowest mass?

[1 mark]

Tick (✓) **one** box.

Electron

☐

Neutron

☐

Proton

☐

0 2 . 5 Neon is unreactive.

Give **one** reason why.

Use **Figure 5**.

[1 mark]

Question 2 continues on the next page

Turn over ►



0 2 . 6 Helium is also in Group 0 of the periodic table.

Give **two** differences between the electronic structure of neon and the electronic structure of helium.

[2 marks]

1 _____

2 _____

0 2 . 7 Some atoms of neon have different numbers of neutrons.

What name is given to atoms of the same element with different numbers of neutrons?

[1 mark]

0 2 . 8 Calculate the mass of 0.42 moles of neon.

Relative atomic mass (A_r): Ne = 20

[2 marks]

Mass = _____ g

11



0 3

This question is about copper and copper compounds.

Copper can be extracted from copper oxide using carbon.

0 3 . 1

Why can carbon be used to extract copper from copper oxide?

[1 mark]Tick (✓) **one** box.

Copper is less reactive than carbon

☐

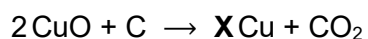
Copper has the same reactivity as carbon

☐

Copper is more reactive than carbon

☐**0 3 . 2**

The symbol equation for a reaction to extract copper is:

What is the value of **X**?**[1 mark]**Tick (✓) **one** box.

1

☐

2

☐

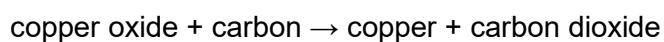
3

☐

4

☐**Question 3 continues on the next page****Turn over ►**

0 3 . 3 The word equation for the reaction is:



Which substance in the equation is reduced?

Give **one** reason for your answer.

Answer in terms of oxygen.

[2 marks]

Substance _____

Reason _____

0 3 . 4 The formula of a different oxide of copper is Cu_2O

Calculate the relative formula mass (M_r) of this oxide of copper.

Relative atomic masses (A_r): Cu = 63.5 O = 16

[2 marks]

Relative formula mass = _____



0	3	.	5
---	---	---	---

A sample of a different copper compound contains:

- 12.7 g of copper
- 5.6 g of nitrogen
- 19.2 g of oxygen.

Calculate the empirical formula of the compound.

Relative atomic masses (A_r): Cu = 63.5 N = 14 O = 16

[4 marks]

Empirical formula = _____

10

Turn over for the next question

Turn over ►



0 4

This question is about energy changes.

A student investigated the change in temperature when three solid substances were dissolved in water.

This is the method used.

- 1 Add 50 cm³ of water into a plastic cup.
- 2 Measure the initial temperature of the water.
- 3 Add 5 g of the solid and start a timer.
- 4 Stir the solution.
- 5 Measure the temperature of the solution after one minute.
- 6 Repeat steps 1 to 5 using different solids.

Table 1 shows the results.

Table 1

Solid	Initial temperature of water in °C	Temperature of solution after one minute in °C
Lithium chloride	19.8	26.7
Magnesium chloride	19.9	35.1
Sodium chloride	19.8	18.6

0 4 . 1

Name **one** piece of equipment that is suitable to measure 50 cm³ of water.

[1 mark]

0 4 . 2

Give **two** control variables the student should keep the same in this investigation.

[2 marks]

1

2



0 4 . 3

The student used a plastic cup instead of a glass beaker.

Suggest why this made the investigation more accurate.

[1 mark]

0 4 . 4

What is the resolution of the thermometer that the student used?

Use **Table 1**.

[1 mark]

_____ °C

0 4 . 5

When **magnesium chloride** dissolves in water the energy change is exothermic.

How do the results in **Table 1** show that the energy change is exothermic?

[1 mark]

0 4 . 6

When **sodium chloride** dissolves in water the energy change is endothermic.

Explain endothermic energy changes in terms of bond breaking and bond making.

[3 marks]



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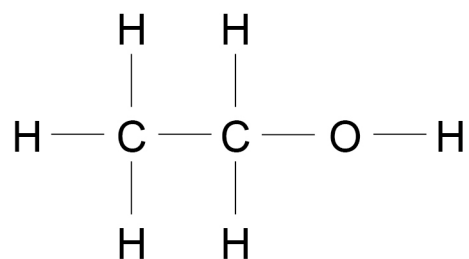
0	5
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This question is about ethanol.

0	5	.	1
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Figure 6 represents the displayed structure of ethanol.

Figure 6



Draw a circle around the functional group in ethanol.

[1 mark]

Question 5 continues on the next page

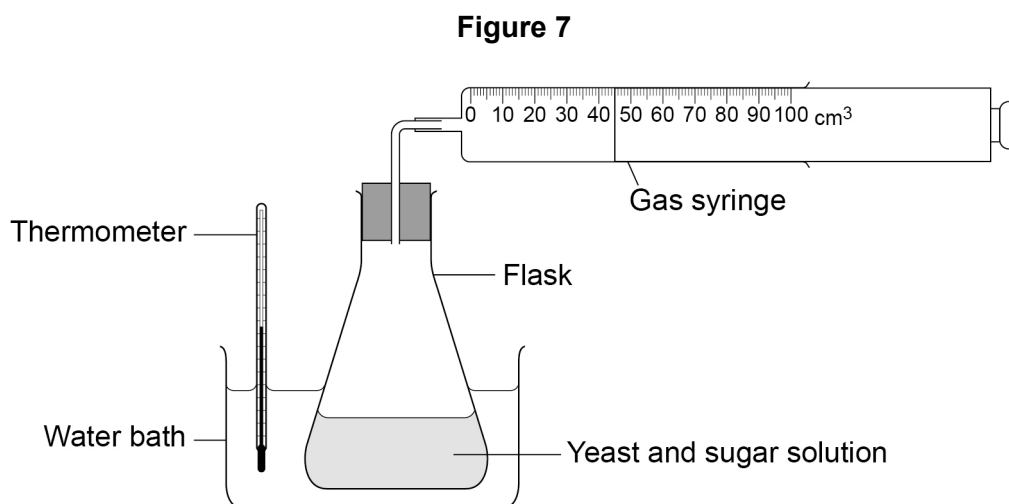


Ethanol can be made by fermenting a yeast and sugar solution.

Carbon dioxide gas is produced in the reaction.

A student investigated the effect of temperature on the rate of fermentation.

Figure 7 shows the apparatus.



The water in the water bath is heated to different temperatures.

The student calculated the rate that carbon dioxide gas was produced at different temperatures.

Table 2 shows the results.

Table 2

Temperature in °C	Rate of production of carbon dioxide in cm ³ /min
20	0.1
25	0.2
30	0.4
35	0.9
40	1.5
45	1.2
50	0.9

0 5 . 2

The student used a water bath instead of a Bunsen burner to heat the yeast and sugar solution.

Give **one** reason why.

[1 mark]

0 5 . 3

Which **best** describes the optimum temperature for fermentation in the investigation?

Use **Table 2**.

[1 mark]

Tick (✓) **one** box.

Optimum temperature < 35 °C

☐

35 °C < Optimum temperature < 45 °C

☐

35 °C > Optimum temperature > 45 °C

☐

Optimum temperature > 45 °C

☐

0 5 . 4

Calculate the volume of carbon dioxide that would be produced in 210 seconds at 40 °C.

Use **Table 2**.

[3 marks]

Volume = _____ cm³

Question 5 continues on the next page

Turn over ►

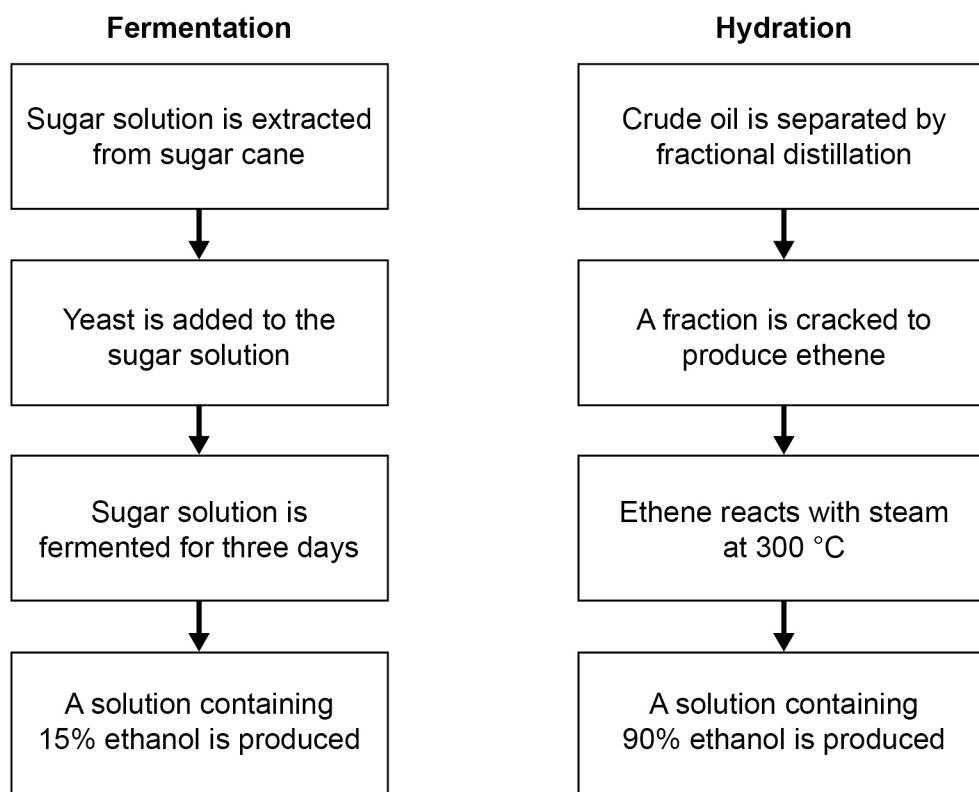


0 5 . 5 Ethanol is produced industrially by fermentation or hydration.

Hydration involves heating ethene with steam at 300 °C.

Figure 8 shows information about the two methods.

Figure 8



You should include economic and environmental factors.

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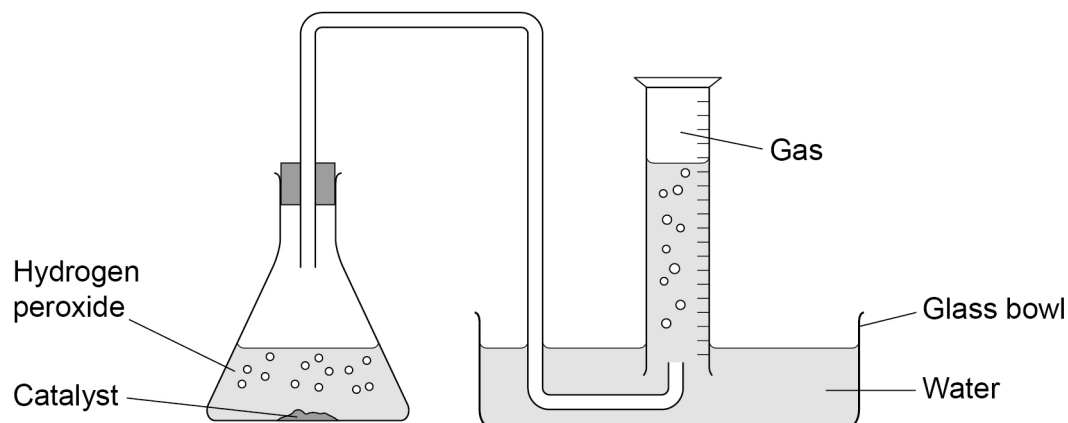
0	6
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A student investigated the rate of decomposition of hydrogen peroxide (H_2O_2) using a catalyst.

Oxygen gas is produced when hydrogen peroxide decomposes.

Figure 9 shows the apparatus.

Figure 9



The student measured the volume of oxygen gas produced every 30 seconds for 5 minutes.

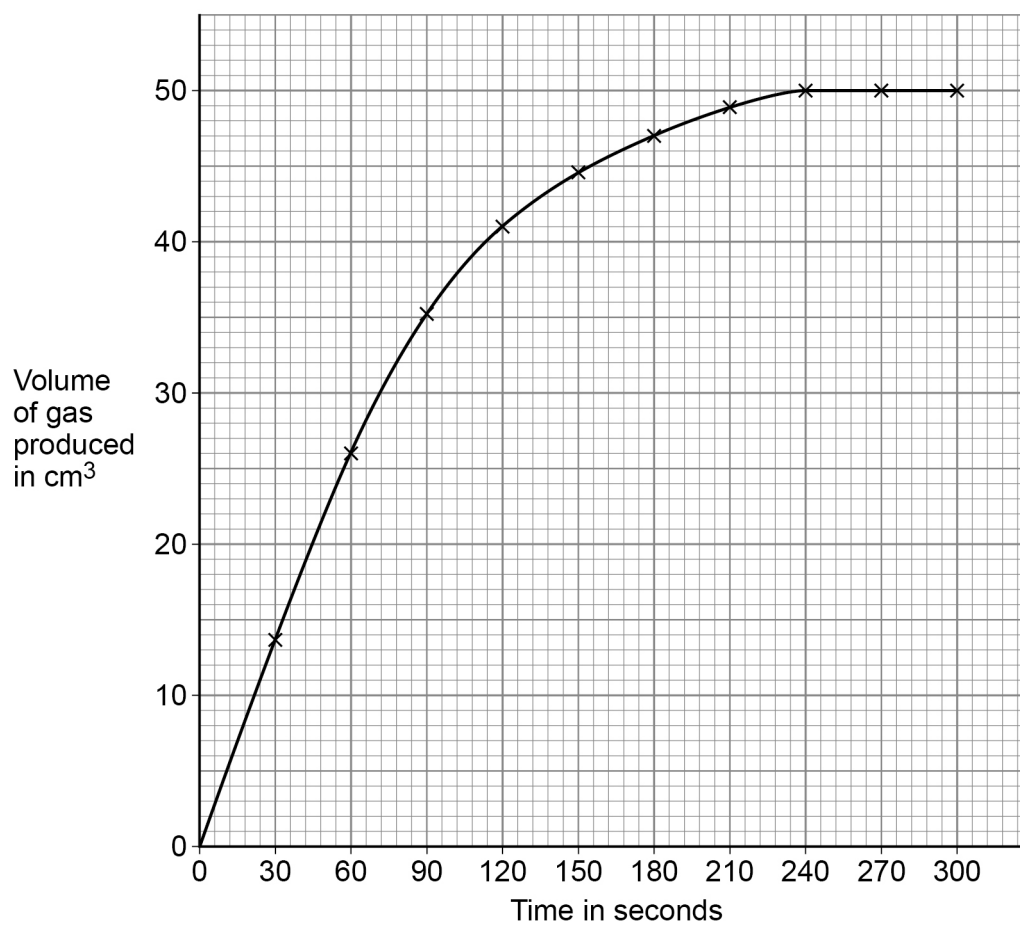
Question 6 continues on the next page

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Figure 10 shows the results.

Figure 10



0 6 . 1 The volume of oxygen the student collected was lower than expected.

Suggest **one** reason why.

[1 mark]



0 6 . 2 Describe the changes in the **rate** of the reaction.

Use **Figure 10**.

[3 marks]

0 6 . 3 Explain why the rate of this reaction changes.

[3 marks]

Question 6 continues on the next page

Turn over ►



0 6 . 4

Table 3 shows the activation energy required for the decomposition of hydrogen peroxide for two different catalysts.

Table 3

Catalyst	Activation energy in kJ/mol
Metal oxide A	33
Metal oxide B	60

The student observed the decomposition of hydrogen peroxide using each catalyst.

Give **one** way in which the observations would be different.

Explain your answer.

[3 marks]

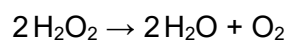
Observation _____

Explanation _____

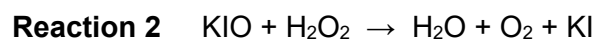
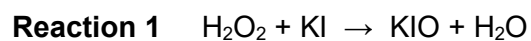


0	6	.	5
---	---	---	---

The equation for the decomposition of hydrogen peroxide is:



Two reactions happen when hydrogen peroxide decomposes with a potassium iodide (KI) catalyst.



Give **two** reasons why these equations show that potassium iodide is a catalyst.

[2 marks]

1 _____

2 _____

Question 6 continues on the next page

Turn over ►



0 6 . 6

Figure 11 shows the displayed structures for the decomposition of hydrogen peroxide.

Figure 11

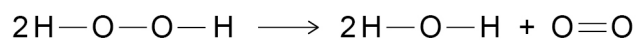


Table 4 shows the bond energies.

Table 4

Bond	O—O	O=O	O—H
Bond energy in kJ/mol	146	496	463

Calculate the energy change of the reaction.

Use **Figure 11** and **Table 4**.

[3 marks]

Energy change = _____ kJ/mol

15



0	7
---	---

This question is about elements in Group 7 of the periodic table.

Group 7 elements form simple molecules.

0	7	.	1
---	---	---	---

The atoms in simple molecules share electrons.

What type of bond is formed when atoms share electrons?

[1 mark]

0	7	.	2
---	---	---	---

Why do simple molecules **not** conduct electricity?

[1 mark]

Question 7 continues on the next page

Turn over ►



Chlorine is in Group 7 of the periodic table.

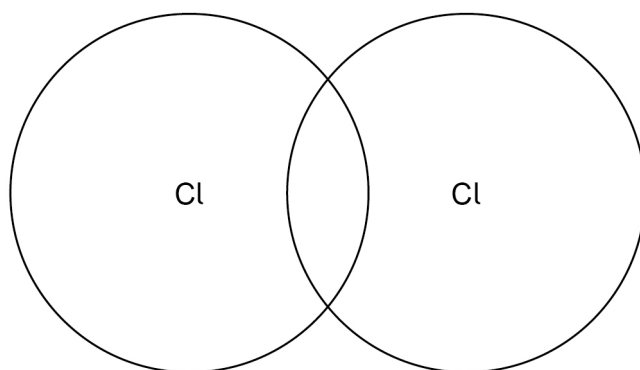
0 7 . 3

Figure 12 shows the outer energy levels in one molecule of chlorine (Cl_2).

Complete **Figure 12** to show the electrons in the outer energy levels.

[2 marks]

Figure 12



0 7 . 4

Damp blue litmus paper is placed in a sample of chlorine gas.

Give **two** changes to the damp blue litmus paper.

[2 marks]

1 _____

2 _____



0 7 . 5 Chlorine atoms are smaller than bromine atoms.

Explain why chlorine is more reactive than bromine.

[2 marks]

0 7 . 6 Chlorine reacts with potassium bromide in a displacement reaction.

Name the **two** products formed when chlorine reacts with potassium bromide.

[2 marks]

1

2

Question 7 continues on the next page

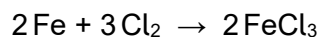
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0	7	.	7
---	---	---	---

Chlorine gas reacts with iron.

The equation for the reaction is:



Calculate the maximum volume of chlorine gas that reacts with 22.4 g of iron.

Give your answer in cm^3 .

The volume of one mole of a gas is 24.0 dm^3 .

Relative atomic mass (A_r): Fe = 56

[5 marks]

Maximum volume = _____ cm^3

15

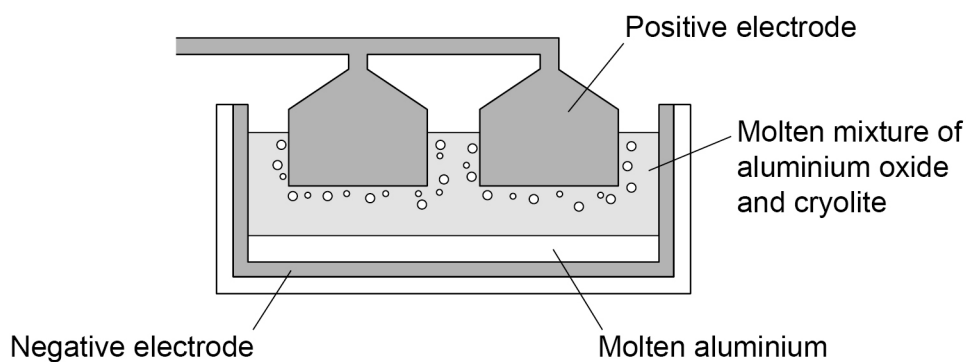


0 8

The electrolysis of aluminium oxide produces aluminium and oxygen.

Figure 13 shows the extraction of aluminium from aluminium oxide using electrolysis.

Figure 13



0 8

1

What is meant by 'electrolysis'?

[2 marks]

0 8

2

Give **one** reason why electrolysis is expensive.

[1 mark]

Question 8 continues on the next page

Turn over ►



0 8 . 3

Why must the mixture of aluminium oxide and cryolite be molten to conduct electricity?

[1 mark]

0 8 . 4

Why is cryolite added to aluminium oxide?

[1 mark]

0 8 . 5

Explain why the positive electrodes must be replaced regularly.

[3 marks]

0 8 . 6

Write the half equation for the production of aluminium at the negative electrode.

[2 marks]



10

END OF QUESTIONS



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